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A systematic review on machine learning deep learning and IoT for water quality prediction

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Abstract

Water quality degradation driven by population growth, industrialization, and climate change poses significant risks to public health and aquatic ecosystems, necessitating accurate and timely prediction frameworks. This systematic review synthesizes recent advances in Water Quality Prediction (WQP) using Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT) technologies. Following PRISMA guidelines, 289 records published between 2020 and 2025 were retrieved from IEEE Xplore, ScienceDirect, MDPI, and Google Scholar, of which 81 studies met the inclusion criteria. The reviewed literature spans diverse datasets from rivers, lakes, reservoirs, and drinking water systems, with predicted parameters including dissolved oxygen, pH, turbidity, biochemical oxygen demand, and water quality indices. In studies, most models report high predictive performance, although the performance metrics vary across datasets and environmental contexts. DL architecture is increasingly adopted for spatiotemporal modeling and large-scale data integration. IoT-enabled systems are employed for real-time monitoring and early warning, in conjunction with ML/DL models. This study emphasizes the role of ML, DL and IoT in supporting sustainable water resource management by enabling accurate real time WQP and contributing to global sustainability goals including SDG 6 Clean Water and Sanitation. Future research should focus on Edge AI, Federated Learning and explainable AI, models to improve secure, reliable, adaptive, and scalable WQP systems.

Keywords Deep learning, Internet of things, Machine learning, Prediction, Water quality

1 Introduction

Water, one of the fundamental resources essential for sustaining life on Earth, and it plays a crucial role for economic growth, human wellness and ecological stability [1, 2]. With the increasing population, urbanization, and climate change the water's quality has significantly decreased [3, 4]. Water contamination has drastic implications, as it can trigger the proliferation of waterborne diseases cholera, dysentery, and typhoid fever, which pose serious risks to human health [5–7]. Therefore, monitoring and maintaining water quality is critical for human, animals and environmental health.

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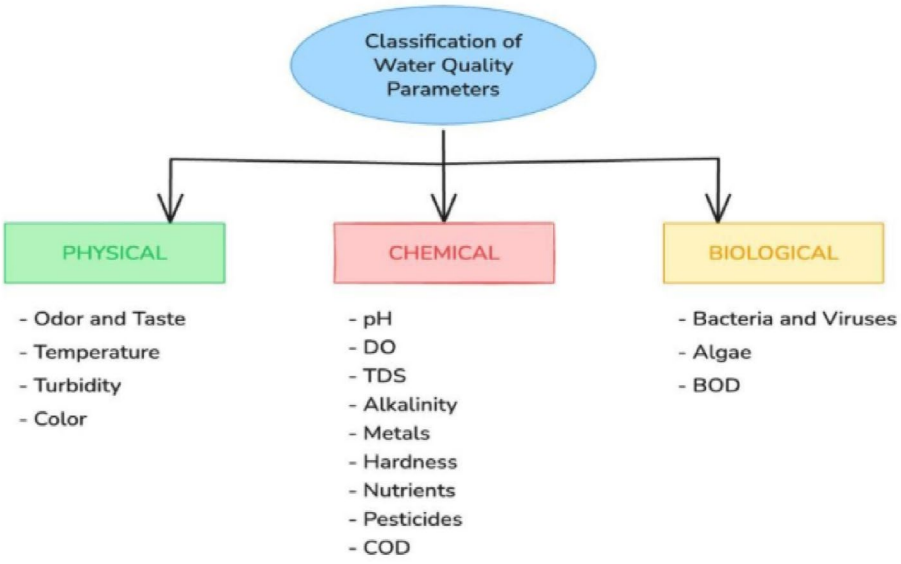


Fig. 1 Categorization of key water quality parameters commonly used in predictive modelling studies

Table 1 Water quality parameters with recommended ranges and measurement units

Parameter	Suitable range	Unit	Refs.
pH	6.5–8.5	–	[13, 29]
Colour	< 15	PCU	[30]
Turbidity	< 5	NTU	[13, 29]
Temperature	< 30	°C	[5]
Nutrients	Water type and pollution level	mg/L	
Hardness	< 60	mg/L as $CaCO_3$	[31]
Dissolved oxygen (DO)	5–8	mg/L	[32, 33]
Alkalinity	20–200	mg/L as $CaCO_3$	
Biochemical oxygen demand (BOD)	< 3(Drinking), < 5(Aquatic Life)	mg/L	[32]
Chemical oxygen demand (COD)	< 10 (Drinking Water)	mg/L	[13, 29]
Total dissolved solids (TDS)	300–600	mg/L	[13, 29]

1.1 Water quality parameters

Water quality (WQ) is the condition of water that determines its suitability for different uses and if it supports aquatic life [8]. It includes the chemical, physical, and biological characteristics of water along with the existence of contaminants and pollutants [9]. By analyzing these characteristics it is possible to understand how effectively WQ meets the requirements of both humans and ecosystems [10] which is shown in Fig. 1.

1.1.1 Physical characteristics

1.1.1.1 Colour It can indicate contamination or the presence of dissolved substances.

1.1.1.2 Turbidity Represents water's degree of clarity and turbidity [11, 12]. Turbidity is assessed to identify the type and extent of treatment required, World Health Organization (WHO) advises that drinking water should have less than 5 NTU to be considered safe for consumption [13–15] which is shown in Table 1.

1.1.1.3 Temperature (TEMP) In WQ assessments, temperature affects solubility, stability, and aquatic life [16]. The water temperature should remain within an appropriate range. Based on WHO guidelines, water used for drinking should have a temperature less than 30 degrees [5, 17, 18].

1.1.1.4 Taste AND Odour It indicates potential contamination which influences the WQ and its acceptability for household and recreational use [19].

1.1.2 Chemical characteristics

1.1.2.1 pH It indicates whether the water is acidic or basic in nature [11]. The recommended pH range for potable water is usually thought to be between 6.5 and 8.5 [13, 20, 21].

1.1.2.2 Alkalinity A water body's buffering capacity, which refers to its capacity to neutralize acids and bases and so preserve stable pH levels.

1.1.2.3 Hardness It affects the use of water and is mostly caused by calcium and magnesium ions.

1.1.2.4 Nutrients In WQ nutrients are essential elements like nitrogen and phosphorus that become pollutants when in excess often from human activities leading to eutrophication. The range of nutrients in water varies with water type and pollution level. For good-quality water, nutrient concentrations are generally low, with nitrate levels typically below 1 mg/L, ammonia less than 0.1 mg/L, and phosphate concentrations usually below 0.05 mg/L.

1.1.2.5 Metals The suitability of water for drinking, irrigation, and aquatic life is influenced by the presence and concentration of metallic elements.

1.1.2.6 Pesticides It is used to protect crops from insects and its contamination in water can risk health of humans and animals consuming it.

1.1.2.7 Total dissolved solids (TDS) Water containing less than 600 mg/L of Total Dissolved Solids (TDS) is considered suitable for drinking, while TDS levels exceeding 1000 mg/L can adversely affect water taste and overall quality. High TDS levels can reduce water acceptability due to increased scaling and mineral deposits in plumbing systems, heaters, boilers, and household appliances. [13, 14].

1.1.2.8 Dissolved oxygen (DO) Indicates the concentration of oxygen within the water [11, 22]. The amount of DO in water is affected by its source, the temperature of the raw water, the treatment it undergoes, and the chemical or biological processes occurring within the distribution system [14]. A good range of Dissolved Oxygen (DO) between 5–8 mg/L is generally considered suitable for surface water bodies and the sustenance of healthy aquatic life.

Reduced levels of DO can harm marine organisms and lead to the degradation of ecosystems, often signalling pollution [23–25].

1.1.2.9 Chemical oxygen demand (COD) The amount of oxygen needed to chemically oxidize organic materials in water with a strong oxidizer is called COD. Microbial breakdown of organic contaminants in high COD levels depletes oxygen, harming aquatic life. COD findings are usually available within 5 h, however BOD testing takes 5 days. Water COD fluctuates greatly based on pollution level and source. COD levels in drinking water are usually below 10 mg/L, while unpolluted surface waters like rivers and lakes are 10–20 mg/L.

1.1.3 Biological characteristics

1.1.3.1 Viruses and bacteria Pathogens that pose health risks are typically managed by water treatments.

1.1.3.2 Algae and other microbes These might be naturally occurring or introduced through contamination; their presence and type can indicate nutrient levels and water health.

1.1.3.3 Biological oxygen demand (BOD) BOD indicates the levels of organic pollution. These characteristics can cause waterborne diseases such as cholera, typhoid fever, Escherichia coli (E. coli), and Salmonella spp. [23, 26–28].

Water quality is largely determined by factors like pH, DO and TDS, as Table 1 illustrates.

Variations from the suggested ranges could be a sign of eutrophication, pollution or inadequate circumstances for aquatic life. A quick reference framework that connects numerical quantities to environmental significance is offered by this tabular depiction.

1.2 Overview of ML, DL and IoT in water quality prediction

Water quality degradation is a major challenge to sustainability and public health and requires accurate prediction for effective monitoring and decision making. Machine learning, deep learning, and IoT provide data driven solutions that enable sustainable water management aligned with SDG 6 Clean Water and Sanitation.

With increasing emphasis on real-time, scalable water quality monitoring, emerging technologies such as ML, DL and IoT are transforming the way water quality parameters are collected, processed and predicted. These tools offer the ability to analyze large volumes of environmental data with enhanced accuracy and minimal human intervention.

ML refers to a class of algorithms that can learn patterns from existing data to make predictions or decisions without being explicitly programmed. Traditional ML models like SVM, DT and KNN are often used for classification or regression tasks, rely on manually selected features from the data. DL is a subset of ML that uses artificial neural networks with multiple hidden layers to automatically extract features and model complex, non-linear relationships. Architectures such as CNN (Convolution Neural Network) and RNN (Recurrent Neural Network) are effective for processing multidimensional or time series water data in an end-to-end learning pipeline. IoT in this context refers to sensor-based systems where interconnected devices continuously monitor physical, chemical, or biological water parameters and transmit the data wirelessly to cloud platforms for analytics. Essential components include water quality sensors, microcontrollers like ESP32, Raspberry Pi, and communication modules like Wi-Fi, LoRa, Zigbee, MQTT.

Together these technologies create an integrated workflow that enables real-time, scalable water quality monitoring and analysis, from data collection through IoT to data processing and prediction using ML/DL models.

2 Literature review

SDG 3 and SDG 12 drive research in water quality monitoring.

Ongoing developments in water quality prediction demonstrate growing sophistication in the selection of parameters and modelling techniques. As summarized in Table 2, the current studies cover several physical characteristics like temperature, pH, conductivity and turbidity. The chemical characteristics like DO, TN, TP, BOD, COD mentioned in [11, 34, 35] and biological markers such as Water Quality Indicator (WQI) [36–38]. This shows the increasing understanding that checking several interconnected parameters is necessary for a evaluation of water quality.

A consistent theme in current studies is the deployment of ML, DL and IoT sensor networks to enable real-time, multi-parameter monitoring and predictive modelling. Many studies use sensor data to continuously measure available parameters while advanced

Table 2 Water quality parameters predicted in recent studies

Refs.	Year	Parameters predicted
[4]	2024	EC, organic carbon, dissolved solids, sulphate, chloramines, pH, hardness, THM levels, turbidity, Drinking suitability
[11]	2025	Temp, DO, pH, turbidity
[47]	2023	Hardness, pH EC, Sulphate, Chloramines, Solids (TDS), Carbon Organic, Turbidity, THs
[48]	2024	pH, Mn, Ni
[49]	2025	NH4-N, BOD, TN, COD
[50]	2024	Chl-a, Turbidity, CDOM, TSS
[51]	2024	DIN, Chl-a, NH4-N
[52]	2024	TN, TP, Chl-a
[53]	2021	BOD, temp, TDS, TSS, turbidity, pH, EC, Alk, Ca, COD, SO_4
[36]	2024	DO, BOD, COD, TSS, EC, pH, temp, Salinity (SAL), Turbidity, DS, NH_4F, NO_3, K, E. coli
[54]	2024	Turbidity, Specific Conductance (SC)
[41]	2022	SAL, pH, temp, DO
[55]	2024	Ni, P, Cl
[56]	2023	Temp, NH_3–N, DO
[35]	2024	DO, TDS, temp, TP, pH, Chl-a, TN, Bacteria, COD, Turbidity, Salinity, NH_3–N
[57]	2024	Temp, DO, AN, TP, TN, Cond, NTU
[58]	2024	pH, DO, AN, TP
[59]	2024	TP, BOD
[34]	2024	DO
[38]	2022	WQI
[37]	2022	WQI
[42]	2022	DO, temp, pH, Cond, turbidity
[46]	2025	DO, nitrate, fecal coliform count, EC, BOD, pH, total coliform count
[39]	2025	DO
[60]	2023	temp
[40]	2020	DO
[44]	2020	pH, temp, DO
[45]	2023	WQI
[61]	2024	P, K, pH, temp, BOD
[62]	2022	NO_3-N, BOD_5, PO_4, NH_3-N
[43]	2023	Temp, DO, pH, ORP
[63]	2023	pH, DO, Temp, EC, ORP
[64]	2022	Temp, pH, DO, BOD, TDS, NO_3

algorithms are used to estimate more complex parameters. These methods are applied in various areas, including drinking water safety, aquaculture and the prediction of WQ in lakes, rivers and coastal regions as well as in water potability.

With this context, DO has emerged as a central focus due to its ecological importance, with many studies applying advanced models such as LSTM, KIG-ELM, FSGCN and CNN to improve its accuracy [39–41]. Similarly, pH, temperature and turbidity have continuously been addressed due to their impact on how chemicals dissolve and how biological processes occur [42–44].

Furthermore, WQI-based frameworks demonstrate the movement toward decision-oriented modeling where models like Gated-LNN and 1D-CNN and MLP are used [37, 41, 45, 46]. Although these models make it easier for policymakers to understand, their dependence on region-specific data raises the possibility that more universal frameworks that can be adjusted to different water systems are required. Overall, literature has evolved from single-parameter models to scalable multi-parameter frameworks for accurate prediction and sustainable water management.

However, despite these advancements, several challenges and research gaps remain. Many studies still rely on a narrow range of parameters typically pH, DO, BOD, COD, turbidity and TDS without incorporating broader indicators that could enhance predictive robustness. The integration of ML, DL, and IoT has enhanced real-time water quality monitoring; however, challenges such as heterogeneous sensor data, unreliable data transmission, and limited adaptability to dynamic streaming data still hinder practical deployment. Additionally, aspects like latency, data security and the infrastructural demands of large-scale monitoring systems receive limited attention. Addressing these limitations will require standardized datasets, adaptive learning mechanisms and secure, edge-enabled architectures that can ensure accuracy, responsiveness, and scalability. By consolidating and analyzing these insights, this review establishes a comprehensive foundation for advancing toward more robust, interpretable and interoperable Water Quality Prediction (WQP) frameworks.

The most frequently used keywords across the reviewed literature are shown in Fig. 2

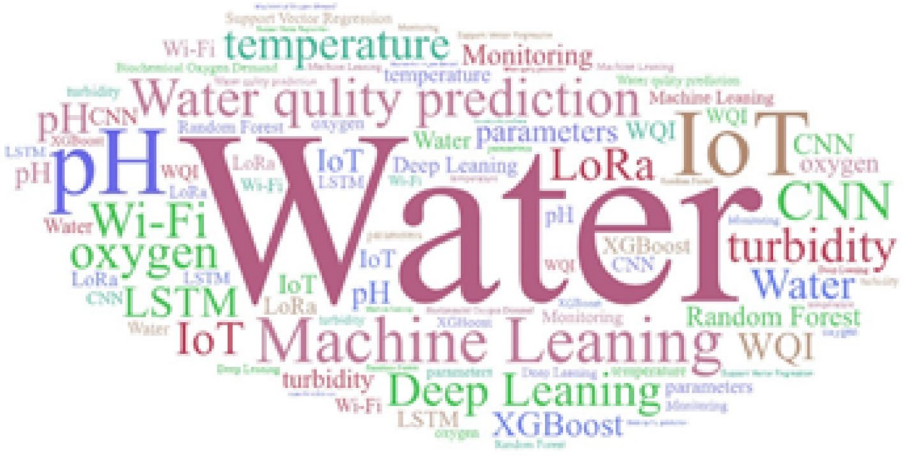


Fig. 2 Word cloud including the most used keywords across the paper

2.1 Motivation

Identifying and addressing possible Water Quality (WQ) problems at an early stage is essential for individual safety and well-being [65]. This is where the prediction of water quality with the help of modern computational techniques becomes essential for accurate monitoring, early contamination detection, and effective water resource management. Numerous studies have explored the application of modern computational techniques in WQP, leveraging their ability to identify complex patterns in water quality data and generate accurate forecasts [4]. Traditional techniques for evaluating water quality mainly depend on physical sampling and chemical analyses conducted in laboratories, which can be slow, require long hours of labor and have restricted spatial and temporal reach [66]. Additionally, they do not meet the need for real-time monitoring of various and diverse WQ parameters, nor do they fulfill the need for highly accurate detection of the factors affecting water quality [47, 67].

In recent years, advancements in technologies such as ML, DL and the IoT have predicted WQ fast and conveniently [68]. The combination of IoT with ML/DL models enables automated prediction systems, which greatly minimizes the need for human involvement and enhances the ability to respond to contamination incidents [65].

While several works cover individual domains of ML, DL and IoT systems, there is a lack of systematic reviews that include all three domains together. This review addresses this gap by critically analyzing current studies, identifying limitations in prior reviews and providing a unified perspective on how ML, DL and IoT are jointly leveraged for multi-parameter WQP. It highlights technological trends, outlines open research questions and demonstrates how these integrated approaches collectively enable real-time, scalable and accurate water quality monitoring, offering guidance for future advancements.

2.2 Contribution

- This review presents a unified cross-domain synthesis of ML, DL, and IoT-based approaches for WQP, bringing together modeling techniques, sensing infrastructure, and deployment architectures within one consolidated review framework.
- In this article provide a structured organization of prior studies based on water quality parameters, prediction objectives (parameter estimation, classification, WQI), and system perspective (data-driven models versus IoT-enabled real-time monitoring), enabling clearer understanding of how these domains intersect.
- The paper consolidates shared challenges, limitations, and open research gaps across all three domains and outlines future research directions for scalable, adaptive, and real-time water quality monitoring systems.

2.3 Organization of the paper

The body of this review paper (Fig. 3), includes Section I which is the introduction, followed by Section II, which discusses the review of literature, summarizing traditional methods and current progress made by applying ML, DL and IoT-based systems for the prediction of WQ. Section III outlines the methodology of this review as per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. This section also highlights how existing works contribute towards answering the defined research questions (RQs). Section IV addresses a discussion on this review

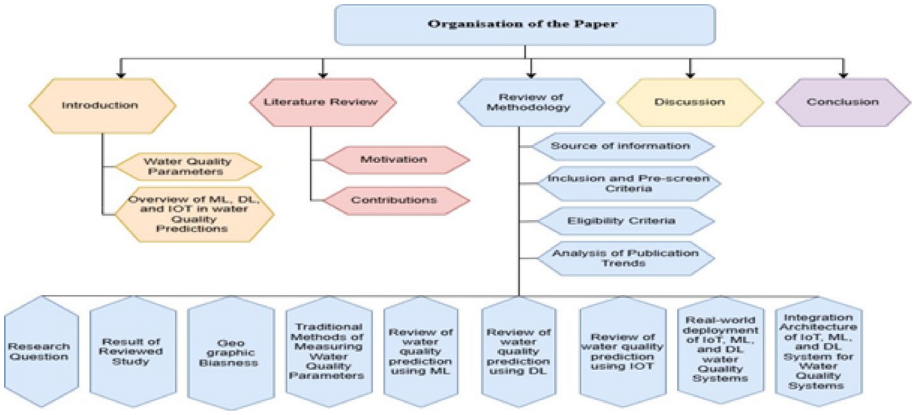


Fig. 3 Organization of this paper

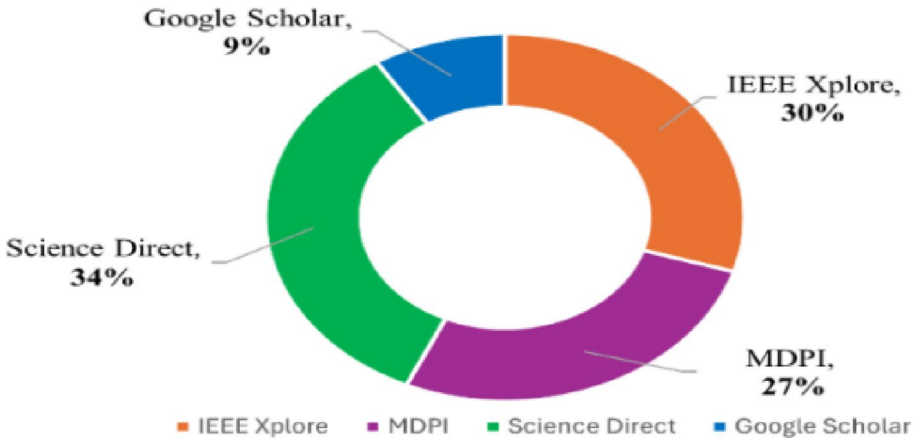


Fig. 4 Distribution of selected articles across databases

paper, presenting the challenges and outlining the future scope in this field. Finally, Section V concludes the study, summarizing the findings of this paper.

3 Review methodology

A systematic review was performed following the standards outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [69]. The criteria for eligibility and exclusion are outlined below:

3.1 Sources of information

To obtain a broad and diverse selection of relevant studies on WQP using ML, DL and IoT, this review examined four primary online databases: 1) IEEE Xplore, which provides research from an engineering and technology perspective, 2) MDPI, known for its vast collection of open-access journals in science and environmental research, 3) ScienceDirect, which features peer-reviewed articles across various scientific fields and 4) Google Scholar, which covers a broad spectrum of academic papers, conference proceedings, and grey literature. This database selection enabled the gathering of a wide range of studies to predict WQ as shown in Fig. 4.

3.2 Inclusion and pre-screening criteria

The chosen articles span publications from 2020 through 2025 to encompass recent developments in the area. Only full-text articles in English were considered for inclusion. Other types of publications, such as unpublished working papers, dissertations, and book chapters, were omitted unless they offered notable methodological advancements. The keyword string used were shown in Table 3. The keyword search yielded an aggregate of 289 articles, 86, 78, 99, and 26 from multiple databases, respectively. Out of the total articles, 33 duplicates were removed, and 256 unique articles were left. These articles were then subsequently screened by examining their titles as well as their abstracts to evaluate their relevance to the subject of predicting water quality with the help of ML, DL and IoT.

3.3 Eligibility criteria

Articles that were pre-screened underwent full-text evaluation, after which relevant and available articles were included and unrelated works were removed. For example, although many retrieved studies applied ML, DL and IoT in fields like agriculture or hospital equipment monitoring, these were excluded as they were not related to WQP. Important information such as key parameters, methodologies and results was examined and then reviewed.

3.4 Analysis of publication trends

Figure 5 shows the year of publication graph of studies published between the years 2020 to 2025. The number of publications has increased regularly, showcasing the rising interest in WQPs through these advanced technologies. The journal analysis identified the top ten publication venues as shown in Table 4 alongside their 2023 impact factors, illustrating the diversity and quality of the publication venues represented in this review. This selection aims to inform readers rather than prioritize or exclude any journals, as all eligible studies were included regardless of their publication venue's impact factor.

3.4.1 Results

Following the full-text evaluation, 175 articles were deemed irrelevant, leaving 81 studies from the original 289 articles. These selected journal and conference papers were examined in detail to investigate the progression and application of ML algorithms for predicting WQ. The review emphasized analyzing different models and assessing their predictive accuracy, operational efficiency, and suitability for various aquatic environments. Significant findings from these studies have been organized and discussed in the

Table 3 Keyword Combinations used for article selection from different databases

S. No	Database	Keyword Combination
1.	IEEE	(("Machine Learning" OR "ML") AND ("Water Quality Prediction" OR "Water Quality Analysis")) AND ("Deep Learning") AND ("IoT")
2.	MDPI	(("water quality prediction")) AND ("machine learning" OR "deep learning") AND ("IoT"))
3.	Google Scholar	("machine learning" OR "ML") AND ("water quality analysis" OR "water quality index") AND ("deep learning" OR "neural network") AND ("internet of things")
4.	ScienceDirect	(("Machine Learning" OR "ML") AND ("Water Quality Prediction" OR "Water Quality Analysis")) AND ("Deep Learning") AND ("IoT" OR "smart water monitoring")

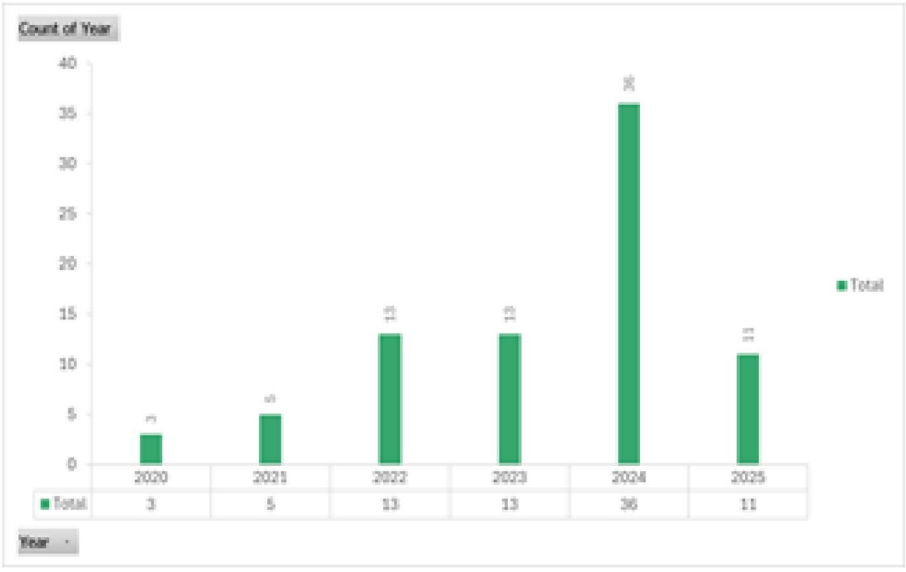


Fig. 5 Number of papers published per year

Table 4 Top ten journals with their impact factor

Journal name	Impact factor (2023)
IEEE Internet of Things Journal	8.2
IEEE Transactions on Geoscience and Remote Sensing	7.5
IEEE Transactions on Automation Science and Engineering	5.9
Environmental Science and Pollution Research	5.8
International Journal of Environmental Research and Public Health	4.6
IEEE Access	3.4
Sustainability (MDPI)	3.3
Water (MDPI)	3.0
Processes	2.8
Water Supply	1.9

subsequent sections. Figure 6 depicts the PRISMA flow diagram [70], that summarizes the selection process and the results of the keyword search.

3.4.2 Research questions

Four principal research questions (RQs) were established to outline the focus of this review. Each question is tackled in its respective section throughout the paper.

1) RQ1: In what ways do ML techniques contribute to water quality prediction? (see subsection I)
2) RQ2: How have DL methods been applied to water quality prediction? (see subsection J)
3) RQ3: How is IoT integrated into water quality monitoring and prediction? (see subsection K)
4) RQ4: What are the challenges of implementing real-time water quality prediction systems? (see section IV and subsection G).

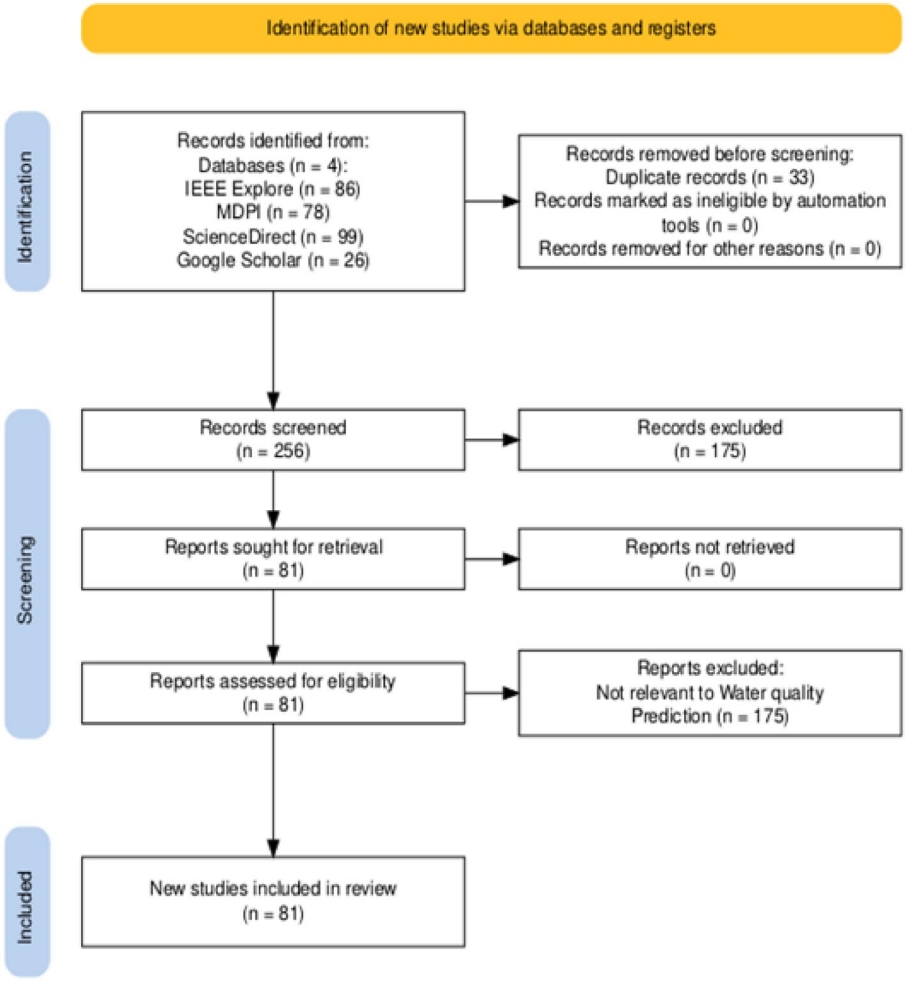


Fig. 6 PRISMA flow diagram

3.4.3 Results of reviewed studies

In Table 5, the dataset size is illustrated, indicating that when the dataset is small, the model fails to detect patterns and captures noise. When small datasets are split, it will directly affect the variance of the results [48]. Small samples also lead to unstable and unreliable performance metrics. Although the model may achieve high accuracy on a small dataset, it often performs low accuracy on new datasets, revealing generalization issues. Additionally, overfitting becomes a significant concern when the dataset is limited, so incorporating synthetic data is advisable. Conversely, when the dataset size is large, it yields stable and reliable performance metrics, reduces overfitting [34], and enhances the model's robustness.

3.4.4 Geographic biasness

Most datasets summarized in Table 5 reveal geographic concentration, with a proportion originating from China [34, 40, 44, 51, 57, 60] and selected regions in the United States [39, 54, 74, 75] and Australia [42]. In contrast, datasets from water-stressed and underrepresented regions including Africa Ethiopia [15], South Asia India [46, 61, 64, 72], and Latin America like Colombia [11] are sparse and limited in scale, or parameter diversity.

Table 5 Overview of water quality datasets used in recent studies

References	Dataset Name	Location	Parameters	Dataset size (sample)	Water body type
[44]	Xincun Marine Aquaculture IoT Dataset	China	Temp, DO, pH	23,204	Marine Aquaculture
[58]	WQ dataset	Jizhou District, Tianjin, China	pH, DO, AN, TP	2,190	River
[48]	Groundwater dataset	Yopurga landfill, China	COD, pH, NH_3-N, Mn, Ni, Fe, B, Pb, Zn, F	29	Groundwater
[71]	WQ and River flow dataset	India and previous papers	pH, hardness, TDS, temp, sulphate	100–1000	River
[26]	Kaggle Drinking Water Quality Classification Dataset	Not specified	pH, hardness, solids, chloramine, sulphates, EC, turbidity, potability	2,000 +	Mixed/ Potability focused
[40]	Real-time water quality and weather datasets	China (Wuxi & Changshu)	DO, pH, temp, air temp, humidity, CO_2, wind, PAR, pressure,	Real-time (288–1008) Day and night	Aquaculture ponds
[60]	Intensive Aquaculture Dataset	Nansha, China	Temp, DO, pH	4,659	Aquaculture ponds
[15]	Drinking water storage tanks dataset	Jimma town, Ethiopia	Temp, other parameters	Samples from 6 tanks TC and FC count 2.6—14.3 CFU/100 and 1.3—9.33 CFU/100	Storage tanks (tap water)
[39]	WQ datasets	USA(Alabama), China (Beijing-Tianjin-Hebei)	DO	19,863 and 7,102	River
[41]	ADAK and MAC WQ dataset	ADAK: Kerala, India; MAC: Hainan, China	SAL, pH, DO, Temp	23,200	Costal Aquaculture zones
[34]	Pearl River Estuary DO Dataset	Dongjiang River, Guangdong, China	DO, pH, Temp, EC, turbidity, COD, TN, TP, heavy metals	34,756 raw records; 8,667 preprocessed	River Estuary
[64]	Ghataprabha River Water Quality Dataset	Karnataka, India	Temp, pH, DO, EC, BOD, TDS	Seasonal (summer, monsoon, winter) 80–200	River
[57]	Qingdao Water Quality Dataset	China	Temp, pH, DO, EC, turbidity, COD, AN, TP, TN	29,448	Surface Water
[54]	In Situ Water Quality Dataset	Lake Houston, USA	Turbidity, EC	Real -time data taken every 15 min (100–1000)	Surface water
[46]	Indian Water Quality Dataset	India	pH, Total Coliform, EC, BOD, Nitrate, DO, Fecal Coliform	1897	Mixed- rivers, lakes etc
[73]	China surface and Bangladesh groundwater dataset	China, Bangladesh	pH, DO, COD_{Mn}, NH_3-N, Temp, Eh, EC, SAL, TDS, Arsenic, anions, cations, trace elements	34,607 (China), 69 (Bangladesh)	Surface water and groundwater
[61]	GIETU Agricultural Fields Dataset	GIETU agricultural fields, India	P, K, pH, Temp, BOD	6 months of sensor data 4324	Reservoirs, irrigation canals, ponds

Table 5 (continued)

References	Dataset Name	Location	Parameters	Dataset size (sample)	Water body type
[11]	Aquaculture Water Quality Dataset	Montería, Córdoba, Colombia	Temp, DO, pH, turbidity	4383	Aquaculture ponds
[42]	Burnett River Dataset (Dataset A) and Baffle River Dataset (Dataset B)	Queensland, Australia	Temp, pH, DO, EC, turbidity	Dataset A: 52,560, Dataset A: 52,560	River (Freshwater)
[74]	Alabama, Beijing & BTH Water Quality Datasets	Alabama (USA), Beijing & BTH (China)	DO, TN	Alabama: hourly DO, Beijing: TN, BTH: 9275 samples	RIver
[75]	Texas Groundwater Database	Texas, USA	TDS	557	Groundwater
[51]	Coastal Water Quality Dataset	Beibu Gulf, China	Temp, pH, Salinity, DO, Chl-a, Nitrate, Nitrite, A, DIN, DIP, plus tidal, meteorological data	10 stations × 10 parameters (2018–2022, half-hourly)	Coastal seawater
[76]	Water Potability Dataset Kaggle	Not specified (Kaggle)	pH, hardness, solids, chloramines, sulphate, EC, turbidity, THMs, potability	935	General drinking water
[45]	Rohri Canal Water Quality Dataset	Rohri Canal, (Sindh, Pakistan)	Temp, pH, Turbidity, TDS	6000, 600	Canal/Surface Water

Models trained predominantly on datasets from China's aquaculture systems or U.S. river basins may learn region-specific climatic conditions which not generalize other environmental contexts. Aquaculture-focused datasets from China [44, 60] highlights parameter like DO, temperature, and pH, whereas groundwater datasets from regions like Ethiopia [15] or Telangana, India [72] involve contamination. Furthermore, datasets from underrepresented regions frequently exhibit limitations in size, sampling frequency, and sensor infrastructure. For example, the groundwater dataset from Yopurga landfill [48] contains only 29 samples, and several Indian datasets [61, 64] rely on short-term observations rather than continuous monitoring. Such disparities can lead to data imbalance and model bias, where ML or DL models achieve high performance on regions but fail when deployed in data-scarce environments.

Another issue is the lack of standardization across datasets, including variations in measured parameters such as COD, BOD, TN, TP vs. minimal parameters like pH and EC, real time and seasonal sampling intervals and water body types.

3.4.5 Traditional methods of measuring water quality parameters

Traditional methods of testing WQ depends vastly on manual sampling and in-lab analysis. These methods, although reliable, are time-consuming, labor-intensive and unfit for providing real-time results. This can result in delayed prediction of pollutants and any changes in water and its timely response for effective administration [13, 77, 78].

In Table 5, the dataset size is illustrated, indicating that when the dataset is small, the model fails to detect patterns and captures noise. When small datasets are split, they

directly affect the variance of the results [48]. Small samples also lead to unstable and unreliable performance metrics. Although the model may achieve high accuracy on a small dataset, it often performs low accuracy on new datasets, revealing generalization issues. Additionally, overfitting becomes a significant concern when the dataset is limited, so incorporating synthetic data is advisable. Conversely, when the dataset size is large, it yields stable and reliable performance metrics, reduces overfitting [34], and enhances the model's robustness.

Traditional testing approaches:

1) Manual and Lab Testing: Samples are collected in person from sites like rivers, lakes, and ground-water that are later sent to labs for testing factors like pH levels, DO and the existence of any chemical components [79].
2) Water Quality Index (WQI) Calculation: WQI [80] is a composite index that results from combining various water quality parameters, each given a weight and sub-index. The final index classifies the water quality as good or poor [48] Traditional approaches to determining water quality indexes are prone to inconsistencies [47].
3) On-site water analysis with the help of tools or instruments where the tool meters parameters like pH, temperature, oxygen levels etc.[81]. This is a quick assessment, but it is limited to some parameters.

Traditional methods are considered good, but they face several limitations including the high cost of continuous sampling and analysis, laborious and time-consuming. With recent advancements and the incorporation of ML, DL and IoT technologies, water quality analysis has improved in efficiency, elaborated in the following sections.

Figure 7 illustrates a typical data workflow where IoT devices collect water parameters and transmit them through communication modules to cloud servers. These data streams are then processed by ML/DL models for forecasting, classification, and real-time monitoring.

4 Review of water quality prediction using machine learning

To address RQ1, this section covers the recent developments in the application of ML [82] models used for WQ prediction. These models are applied to predict various WQ parameters with improved accuracy and scalability. Common algorithms include RF, SVM [83] and DT [84] and other advanced ensemble techniques.

Table 5 provides a summary of the major datasets used in current WQP research. These datasets vary in terms of region, size, parameters and collection methods, providing diverse opportunities and highlighting the heterogeneity in ML applications for WQ prediction.

Various studies have demonstrated the effectiveness of ML models. For instance, in [47], authors have used a supervised ML approach to predict WQ emphasizing that all input parameters are crucial for accurate prediction, with RF outperforming with the highest performance (78% accuracy) in prediction tasks. Similarly, in [48] DT, RF and XGBoost were compared for groundwater quality classification and WQI estimation, with XGBoost showing the most robust performance, especially when using a novel Sinusoidal Weighted Mean (SWM) WQI model, which yielded no overestimation and only 34% underestimation.

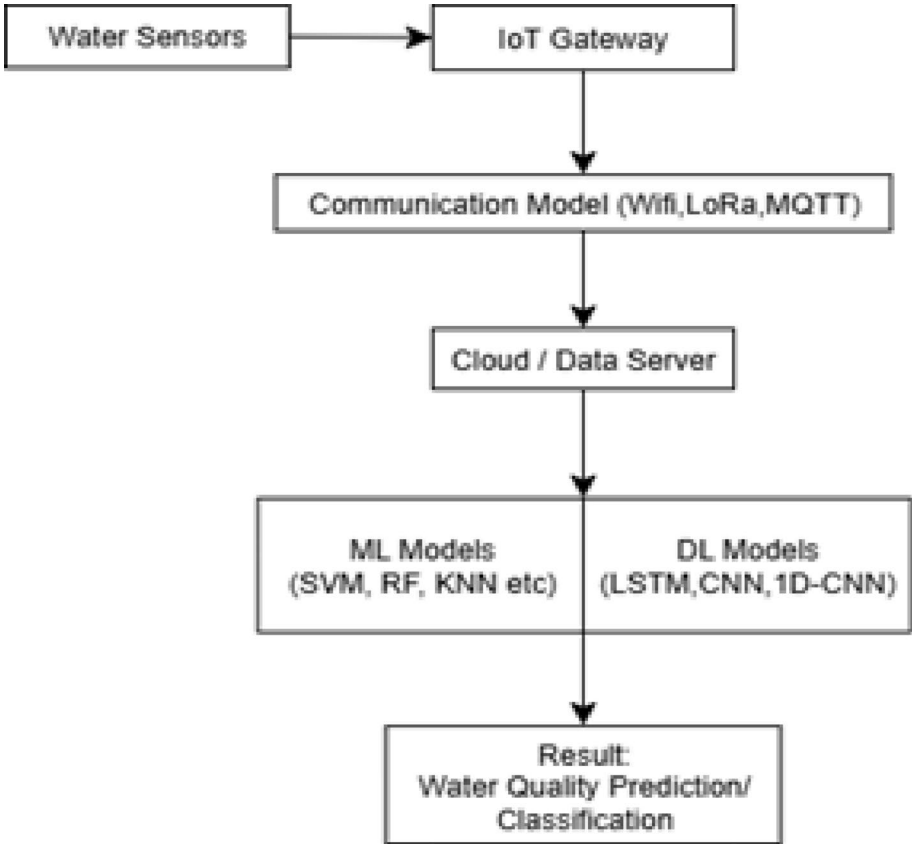


Fig. 7 IoT-based architecture for prediction using ML/DL

Expanding the scope, [49] evaluated seven ML algorithms including LR [85], RF, SVR, KNN, DT, GB and AR models, were applied to predict WQ parameters such as NH4-N, BOD, TN and COD. The GB model achieved the highest overall accuracy $R^2 = 0.75$ while RF demonstrated superior performance specifically for TN and COD predictions.

Similarly, in a comparative study between XGBoost and CatBoost [72], authors found XGBoost to outperform CatBoost by achieving an accuracy of 96.2% compared to CatBoost which achieved 94%. In another comparative study [86] for predicting water potability, authors conducted a of KNN, SVM, DT, and RF. RF showed the best performance before data oversampling 96.64% while DT became the top performing after oversampling with 92.66%.

The integration of ML with remote sensing satellite data was explored in [50], where different satellite platforms were assessed for their suitability in lake WQ monitoring Landsat imagery combined with RF was found effective for large lakes, while Sentinel-2 with GB/MLP and RapidEye with CNN/GAN were suited for medium and small lakes, respectively. Key predicted parameters included Chl-a, CDOM, TSS, and temperature.

Lastly, classification-based approaches were explored in [87], using models like KNN, NB, SVM, DT and RF. RF was identified as the top performer with an accuracy of 91.97%, with pH, DO and nitrate as key predictors. An ensembled ML model made by integrating DT, LR and three RF models was explored in [88], further improving performance by 92% and SVM (85%), RF (87%), DT (81%) and LR (81.6%).

Table 6, summarizes the ML models, their task, algorithms used and their results in recent studies.

TABLE 6 Summary of ML model performance for WQ prediction

Ref., Year	Task	Algorithms Used	Best Performing Algorithm	Parameters Predicted	Pre-processing	Results (in %)	
						Accuracy	R^2
[51], 2024	Coastal Water Quality Monitoring	RF, Linear Regression	RF	Chl-a, DIN, NH4-N	Secondary moving average	60	–
[4], 2024	Water Quality Prediction	DT, SVM, KNN, RF	DT	pH, Hardness, Solids, Sulphate, Chloramines, EC, Organic Carbon, THM, turbidity, Potability	–	62.21	–
[75], 2024	Predicting TDS	DT, RF, KNN	DT	TDS	Removal of null values and duplicates	99.80	–
[52], 2024	Water Quality Monitoring	Stacked-Random Forest (ST-RF), LASSO, MLR, GB, CatBoost, AdaBoost	ST-RF	Chl-a, TN, TP	Geometric and atmospheric correction	–	82.3
[48], 2024	Groundwater Quality near Yopurga landfill	DT, RF, XGBoost-based WQI model with SVM	XGBoost	pH, Mn, Ni	Z-score normalization	87.5	–
[47], 2023	Realtime Water Quality Prediction	LR, KNN, RF, ANN	RF	Hardness, pH, and EC, Sulphate, Chloramines, TDS, Carbon Organic, turbidity, THMs	MinMaxScaler	78	–
[49], 2025	Predicting water quality parameters	LR, KNN, SVR, DT, RF, GB and AdaBoost	GBR	NH4-N, BOD, TN, COD	Linear interpolation	–	75
[53], 2021	Surface Water Quality Prediction	QRF, RF, SVM, Stochastic Gradient Boosting (GBM), GA, PCA	PCA-QRF	BOD, temp, turbidity, pH, EC, Alk, Ca, COD, SO_4, TDS, TSS	Truncation selection	–	94
[76], 2023	Water Potability Classification	Proposed AdCSO-sELM (Adaptive Crow Search Optimized SoftMax-Extreme Learning Machine)	AdCSO-sELM	NA	KNN	96.54	
[36], 2024	Classify River Water Quality for Kinta River	DT, LR, RF, SVM, AdaBoost, XGBoost	GridSearchCV ensemble	DO, BOD, COD, TSS, EC, pH, TEMP, SAL, turbidity, DS, NH4F, NO3, K, E. coli	Z-score normalization	97.3	

TABLE 6 (continued)

Ref., Year	Task	Algorithms Used	Best Performing Algorithm	Parameters Predicted	Pre-processing	Results (in %)	
						Accuracy	R^2
[54], 2024	Prediction of Water quality parameters for Lake Houston	Mixture Density Networks, SVR, Partial Least Squares Regression, FSA (Fuzzy Similarity Analysis)	FSA	Turbidity, SC	Log transformation	–	92

In summary, RF consistently showed high predictive accuracy across various environments. Ensemble models, such as ST-RF and GridSearch CV, demonstrated enhanced generalization, with some achieving accuracies above 97%. However, prediction effectiveness varies with water environment, dataset quality, and parameter diversity. Future improvements could focus on feature selection, real-time integration with IoT and model interpretability to ensure broader adoption.

4.1 Review of water quality prediction using deep learning

To answer RQ2, this section reviews recent advances in applying DL models like LSTM, CNN, RNN and GANs for accurate water quality analysis. These models entail acquiring complicated feature representation and recognizing patterns through multi-layered neural networks [66, 89].

The evolution of DL models has led to the development of hybrid models for WQ prediction. For example, a study in [41] proposed a hybrid CNN-LSTM model that utilized two datasets (ADAK and MAC) to assess the effects of hyperparameters. The model exhibited exceptional accuracy and computational efficiency, obtaining a RMSE of 0.3217 (parts per thousand) ppt and MAE of 0.2509 ppt on the ADAK dataset, while the MAC dataset yielded MAE of 0.0381 ppt and RMSE of 0.0827 ppt.

Extending this direction on advanced graph-based architecture, authors in [55], introduced a Triple-Channel Adjacency-Attention Graph Convolutional Network, utilizing reinforcement learning to improve feature extraction. This model showed strong results, with a reported RSE of 0.071 and CORR of 0.928.

Optimization techniques are increasingly being incorporated into neural network designs to improve their effectiveness, which is notable. In [58], an Attention-enhanced LSTM with Adaptive Particle Swarm Optimization (AWPSO-LSTMAT) model presented combines attention mechanisms with adaptive particle swarm optimization, achieving outstanding results for various water quality parameters like pH and DO. This demonstrates how metaheuristic optimization can enhance the performance of DL models in real-time WQ monitoring.

In terms of domain-specific applications, [34] developed a MixUp-LSTM model trained on estuarine data over four years. It accurately predicted DO levels with $R^2 = 0.9208$, though limited data hindered the identification of hypoxia sources (i.e. Hypoxia is a condition where dissolved oxygen levels in water drops too low to support aquatic life). The study emphasizes that domain-specific parameters like pH, NH_3-N, and conductivity significantly influence DO and should be prioritized in future modelling efforts.

When real-time data is scarce DL models trained on historic records prove highly effective. For instance, [38] evaluated both feedforward and recurrent models with LSTM achieving up to 98% accuracy underscoring the viability of memory-based models in resource-limited environments.

Another emerging approach is classification of model structures based on time-series handling as seen in [35]. Here WQ models were categorized into Point-to-Point (P2P), Series-to-Point (S2P) and Series-to-Series (S2S). While P2P models favored traditional ML like RF and SVM, S2S architectures benefited from advanced DL models such as attention-based GRU (AT-GRU) and transformers. This framework aids researchers in selecting the appropriate model type based on the temporal nature of the data.

Despite these advancements, traditional ML models still serve as valuable baselines. For example, in [56] Orange data mining tool were utilized to apply various ML models, including DT, GB, and neural networks (NN), for water safety prediction. NN, GB, and DT yielded top accuracies of 99.8%, 99.2%, and 99.7% respectively, with AUC values reaching 1.000 and recall scores nearing 99.8% showing that in some tasks, classical models may still compete with or complement DL systems. These comparisons help frame DL not as a replacement but as a progressive enhancement built on solid ML foundations.

Table 7 presents an overview of all the DL models used in the prediction of WQ.

In summary, DL models such as CNN-LSTM, MixUp-LSTM, AWPSO-LSTMAT and STGFT have demonstrated superior performance in predicting water quality parameters across diverse aquatic environments. Attention-based and graph-based architectures showed strong capabilities in capturing temporal and spatial dependencies, while hybrid and ensemble strategies improved accuracy and robustness, with several models achieving R^2 values above 0.90. However, DL performance is often contingent on large, high-quality datasets and careful tuning of architecture and hyperparameters. Future advancements may focus on lightweight models for real-time deployment, better interpretability, and tighter integration with IoT and sensor networks to support dynamic, data-driven water quality management.

4.2 Review of water quality prediction using IoT

Addressing RQ3, this section highlights recent advancements in the use of IoT-based architecture for the prediction of water quality. A typical IoT-enabled system begins with sensor nodes that measure parameters such as pH, turbidity, temperature, dissolved oxygen and TDS.

These sensors are connected to microcontrollers like Arduino, ESP32, or Raspberry Pi, which acquire, pre-process and format the data for transmission. Communication modules such as Wi-Fi, Zigbee, LoRa or MQTT facilitate wireless data transfer to the cloud or local servers, where ML and DL algorithms are applied for real-time WQP [5].

This data allows for scalable, remote and continuous monitoring, enabling faster detection of contamination events and improved water resource management. Figure 8 illustrates a typical architecture of an IoT-enabled WQ monitoring.

Several core water quality parameters are monitored using specific sensors:

1) pH Sensors, such as the SEN0161, operate over a range of 0–14 with an accuracy of ± 0.1 pH units and a resolution of 0.01 pH, allowing accurate assessment of water acidity or alkalinity [10, 90].

Table 7 Summary of DL-related studies used in water quality prediction

Ref., Year	Task	Algorithms Used	Best Performing Algorithm	Parameters Predicted	Pre-processing	Results	
						ACCURACY	RMSE
[37], 2022	WQI Prediction of Urban Rivers	RF, 1D-CNN	1D-CNN	WQI	Wavelength calculation and Savitzky-Golay processing	–	0.574
[41], 2022	Aquaculture Water Quality Prediction	Hybrid CNN-LSTM, CNN-GRU model	Hybrid CNN-LSTM	SAL, pH, temp, DO	Minimum Maximum Scaling	–	0.3217
[55], 2024	WQ prediction using reinforcement learning	Multi-channel graph convolution autoencoder	DRL-based triple-channel adjacency attention graph convolution model	Ni, P, Cl	Data split according to time of observation	–	0.071
[56], 2023	WQ Data Prediction of Transboundary Watershed	ARIMA, ST-KNN, ST-GCN, TRMF, GC-DCRNN, GEOMAN, SGL, MDMVL	MDMVL framework	Temp, NH3 -N, DO	Multi variant data multiview model for predicting missing values	–	1.66
[57], 2024	WQ Classification Prediction	DT, KNN, Ensembled Trees (Soft/Hard voting), LR	Soft Voting Ensemble tree algorithm	Temp, DO, AN, TP, TN, EC, turbidity	Gradient-based one-side sampling (GOSS)	99.91%	-
[58], 2024	WQ Prediction in Zhou River	SVR, LSTM, CNN-LSTM, CNN-GRU, AWPSO-LSTMAT	AWPSO-LSTMAT	pH, DO, AN, TP	Linear interpolation	–	0.029
[59], 2024	WQ Prediction using Fuzzy NN	Deep fuzzy Learning (DFNN), Event-Triggered Deep Fuzzy Neural Network (ET-DFNN)	ET-DFNN	TP, BOD	–	–	0.0109
[34], 2024	DO prediction and Identification of its driving factors	MIC, XGBoost, LSTM, MixUp-LSTM	MixUp-LSTM	DO (Parameters causing low DO: pH, EC, NH_3-N)	Nearest interpolation	–	0.4100
[38], 2022	WQ Prediction	ANN (Feedforward, RNN), LSTM	RNN-based LSTM model	WQI	Linear interpolation, smoothing and moving average filtering	96%	0.24
[42], 2022	LSTM Water Quality Prediction	LSTM, Genetic Algorithm (GA)-LSTM, Ensemble model	GA-LSTM	DO, temp, pH, EC, turbidity	first, second, and third quartile was used to remove outliers	–	0.25
[46], 2025	Prediction and classification of WQI	Gated-LNN model (Gated Liquid Neural Network)	Gated-LNN	pH, EC, BOD, nitrate, total coliform, DO, fecal coliform	Z-score normalization	99.74%	0.0357

Table 7 (continued)

Ref., Year	Task	Algorithms Used	Best Performing Algorithm	Parameters Predicted	Pre-processing	Results	
						ACCURACY	RMSE
[39], 2025	Spatiotemporal WQ Prediction	Fusion Spatio-temporal Graph CNN (FSGCN), ARIMA, ANN, Seq2Seq, STGCN, DCRNN, GraphWaveNet,	FSGCN	DO	–	–	0.29
[60], 2023	Prediction of temperature of water in Aquaculture	VMD, CNN, LSTM, GRU, VMDCNN-BiLSTM-SA	VMDCNN-BiLSTM-SA hybrid model	Temp	Three sigma rule	–	0.143
[40], 2020	Prediction of DO	KIG-ELM (K-means, Improved Genetic Algorithm (IGA), Extreme Learning Machine (ELM))	KIG-ELM	DO	Z-score normalization	–	0.2592
[44], 2020	WQ Prediction in Smart Mariculture	RNN, LSTM, Bi-S-SRU (Bi-directional Stacked Simple Recurrent Unit)	Bi-S-SRU	pH, temp, DO	Mean proximity method	94.42%	0.0261



Fig. 8 IoT-based Water Quality Monitoring Architecture

2) Turbidity sensors measure water clarity by detecting the amount of light scattered by suspended particles. Sensors such as the SEN0189 can measure turbidity levels up to 1000 NTU with high accuracy [91].
3) Dissolved Oxygen (DO) Sensors: These sensors measure the amount of oxygen in water, which is important for the survival of fish and other aquatic life, it measures oxygen levels between 0–20 mg/L with a high accuracy [90].
4) Temperature Sensors, like the DS18B20, offer a wide detection range from –55 °C to + 125 °C and an accuracy of ± 0.5 °C, useful for environmental and industrial monitoring [90, 92].

These sensors are typically interfaced with microcontrollers such as Arduino, ESP32 or Raspberry Pi, which process the sensor data and support real-time transmission via various wireless communication modules [90].

Commonly used communication technologies include Wi-Fi, Zigbee, LoRa, Bluetooth, Cellular and MQTT [10, 93]:

1) Wi-Fi: Offers a moderate range (up to 100 m) and fast data transfer rates (up to 150 Mbps), making it suitable for local or indoor monitoring setups.
2) Zigbe: A low-power protocol with a range of 10–100 m, ideal for creating mesh networks in distributed sensor deployments.
3) LoRa (Long Range): Designed for low-power, long-range transmission (2–15 km) at data rates between 0.3–50 kbps, making it well-suited for rural or remote water monitoring applications.

4) Bluetooth: Provides a short communication range (10 m) with low energy consumption, often used in mobile or wearable water quality systems.
5) Cellular (GSM/3G/4G/5G): Provides wide coverage but consumes higher power and incurs operational costs.
6) MQTT (Message Queuing Telemetry Transport): Lightweight publish-subscribe protocol suited for real-time sensor communication over constrained Networks.

For instance, in [45], to anticipate the WQI and Water Quality Class (WQC), an IoT system based on Long Range (LoRa) was developed to gather data on pH, temperature, turbidity and TDS. Similarly, the study [73] suggested a wireless IoT framework using Zigbee modules and AWS services, focusing on the prediction of parameters such as pH, DO, COD and NH3-N through GridSearchCV, for assessing the purity of water in real-time.

Aquaculture focused studies in [11, 43], employ Wi-Fi-connected sensors to gather data on turbidity, DO, pH and temperature. These systems use ML and DL models like Optimization Algorithm (QAOA) and Simple Recurrent Units (SRU) for intelligent analysis and monitoring of the WQ.

Further highlighting the potential of IoT, [5] examined several wireless communication networks, involving LPWAN, Wi-Fi, Zigbee, RFID and Bluetooth, and their integration with supervised and unsupervised ML models. These technologies allow for accurately measuring parameters like temperature, turbidity, DO and pH.

In [94], IoT-based water quality monitoring systems utilize sensors such as pH, turbidity, TDS, DO and temperature to collect real-time data. This data is transmitted via gateways like Raspberry Pi or ESP32 using lightweight protocols such as MQTT or CoAP to cloud platforms for analysis. Integrated with ML frameworks like EWAIS, the system enables accurate and interpretable WQPs. This approach supports timely decision-making and enhances sustainability in urban water management. This growing emphasis on explainability is echoed in [61], which introduced an MQTT-based IoT-AI system to analyze agricultural water, achieving 90% classification accuracy and improved interpretability.

The integration of IoT with Geographic Information Systems (GIS) was explored in [1], where smart sensors connected via LPWAN were used to monitor pH, turbidity, TDS, and temperature. This configuration achieved 95% accuracy, while GIS tools enabled spatial water management and pattern analysis.

Collectively, these studies reflect a broader trend toward automated, intelligent WQ monitoring. As highlighted in [11], emphasized the broader trend of transitioning from manual water testing to automated IoT-ML systems, noting improvements in cost-efficiency, predictive maintenance, and flexible monitoring.

Nonetheless, several challenges persist. These include the lack of standardized protocols across hardware and networks, hardware durability under varying environmental conditions, and ensuring model generalizability across different geographies and water bodies. Addressing these gaps will be essential for building robust, large-scale WQ prediction systems in the future.

Table 8 summarizes recent studies that utilize the IoT framework for WQ assessment.

For long-term applications, water quality system monitoring and testing have adopted various techniques, including Zigbee, LoRa, BLE, and Lora WAN, because these systems rely on batteries. However, the choice of these technologies greatly affects system

Table 8 Summary of IoT-related studies used in water quality prediction

Ref., Year	Task	Communication Model	Parameters Predicted	ML/DL Models Used	Pre-processing	Results	
						Accuracy	RMSE
[45], 2023	IoT-Based Prediction of WQI and WQC	Long-Range (LoRa)	WQI, WQC	MLP, RF	Z-score normalization	91%	13.6
[73], 2024	Real-time WQ prediction	Zigbee, AWS, MQTT, Telegraf	pH, DO, COD, NH_3-N	RF, XGB, HGB	Weighted Arithmetic Index	99.80%	–
[61], 2024	Real-Time IoT and AI-Based Multi-Nutrient WQ Prediction	MQTT Protocol, Cloud	P, K, pH, temp, BOD	RF + SVM, XAI	–	90.99%	–
[11], 2025	Intelligent WQ Prediction and Monitoring in Aquaculture	Wi-Fi	Temp, DO, pH, turbidity	RF, QAOA	Min–Max Scaling	–	0.0998
[62], 2022	Prediction of water quality for freshwater biodiversity	Wi-Fi, 5G	NO_3-N, BOD_5, PO_4, NH_3-N	GRNN, MPR	Centered to the mean and component wise scaled to unit variance	–	0.04 (Mean Avg Error)
[95], 2022	WQ monitoring in Harsh Tropical environment	LoRa, Wi-Fi, SUN	PDR, PL, temp, pH, pressure	Hybrid ML-based semi-empirical path loss model	filtering/cleaning the measurements from outliers and irregular measurement points	90%	3.5
[43], 2023	AIoT-based WQ montoring and prediction system for Aquaculture	Wi-Fi	Temp, DO, pH, ORP	SRU	Moving average processing	–	0.0157
[64], 2022	WQ monitoring (WQM)	transceivers	Temp, pH, DO, BOD, TDS, NO_3	LR, ANOVA	–		0.4472 (Standard Error)
[96], 2024	Agriculture water quality	Wi-FI	pH, TDS, temp, Turbidity	SVM, RF, DT, LR, Naïve Bayes	All symbolic features are translated to numeric features	95%	–
[97], 2024	Water resource monitoring	LoRaWAN	Temp, DO, ORP, and pH	None	–	98% (packet reception rate)	–
[98], 2022	Water quality prediction	Sigfox	pH, EC, oxygen, temp	LSTM	Z-score normalization	–	0.092 (Mean Squared Error)

performance in terms of energy efficiency, cost-effectiveness, scalability, and reliability. Because of their low energy consumption and higher latency, the use of these technologies is limited in real-time monitoring applications [5, 68]. On the other hand, Wi-Fi, 5G, and GSM need a lot of power to transmit data and maintain a connection [10]. Cloud-based platforms, such as AWS, are more expensive because they need continuous data, storage, and maintenance [5].[45]. In contrast to small-area Wi-Fi and Bluetooth, which are utilized for a restricted range, LoRa and Sigfox are used to cover a broad area for the prediction of water quality tests [61]. In terms of cost, WQP based on a small system requires Arduino and ESP systems, which are less expensive compared to other

devices because they don't need continuous internet. Wi-Fi [45] has a higher failure rate, whereas LoRa and LoRa WAN have a low failure rate [68]

Low-power wide-area networks like LoRa, Sigfox are used for large-scale, low-frequency monitoring due to energy efficiency and long-range coverage, but they do not support real-time control. Mesh networks provide reliable communication in distributed sensing environments with moderate latency. Wi-Fi, 5G, MQTT-based cloud systems enable real-time analytics but require higher energy and maintenance, limiting their use in remote deployments.

4.3 Real-world deployments of IOT–ML/DL water quality systems

Table 9 Represents the Communication technologies and network topologies in IoT-Based Water Quality Monitoring Systems. Real-world deployments are demonstrated across communication models, but system capabilities differ by topology and protocol. LoRa/LoRaWAN and Sigfox systems are deployed in canals, agricultural fields, and remote water bodies. These systems achieve long-range coverage (5–15 km) and > 97% packet reception with multi-year battery life. Due to high latency and low bandwidth, they support periodic monitoring only and do not enable real-time control. Zigbee, XBee-PRO, and SUN mesh networks are deployed in multi-node river and distributed sensing setups. These systems provide millisecond-level latency and maintain communication through multi-hop routing, ensuring high reliability under node failure. Their limitation is restricted range (~ 100 m per node) and increased maintenance with network size. Wi-Fi, Wi-Fi + 5G, and MQTT-based cloud systems are implemented in aquaculture and irrigation systems. These systems support real-time data streaming and ML-based prediction with low latency (milliseconds to 1–10 ms for 5G). However, they require continuous power, stable internet connectivity, and regular maintenance, restricting deployment in remote locations. Bluetooth and point-to-point transceivers are used in short-range deployments. They provide low latency and low power operation, but limited coverage and interference sensitivity restrict them to small-scale setups. Across all studies, systems are deployed in field environments and achieve reliable data acquisition and prediction. However, no system implements closed-loop control, autonomous actuation, or large-scale integrated operation. Current implementations function as monitoring and decision-support systems, not fully autonomous water management solutions.

4.4 Integration architectures of IOT–ML/DL systems for water quality prediction

- The reviewed studies were categorized based on the integration of IoT with ML/DL processing pipelines. Studies [64] and [97] fall under IoT-only monitoring. In [64], IoT transceivers are used for real-time data acquisition, and analysis is limited to statistical methods, with ML-based prediction not implemented. In [97], the system performs only data collection and transmission using LoRaWAN, without any ML/DL component. No study demonstrates IoT-only monitoring with offline ML/DL analysis, as none perform ML/DL separately after data collection without system integration.
- Studies [73] and [61] implement IoT + cloud-based ML/DL (centralized processing). In [73], sensor data is transmitted via Zigbee and MQTT to AWS, where preprocessing, storage, and ML inference are executed. In [61], IoT devices transmit

Table 9 Communication technologies and network topologies in IoT-based water quality monitoring systems

Communication model	Network topology	Latency & real-time capability	Energy consumption & maintenance	Reliability & field robustness
LoRa	Star	Latency is high and measured in seconds not suitable for real-time control	Very low power; battery life up to years; minimal maintenance	High robustness over 5–15 km; performs well in rural and areas
Zigbee	Mesh	Latency is low and measured in milliseconds; supports near real-time sensing	Low power; moderate maintenance due to multi-node routing	High reliability due mesh network; ~ 100 m range per node
MQTT	Publish–Subscribe (Broker-based)model	Low latency if network stable; supports real-time streaming via cloud	Depends on Wi-fi or cellular network; moderate energy use	Highly reliable with Quality of Service (QoS) levels; depends on internet connectivity
Wi-Fi	Star	Latency measured in milliseconds; supports real-time monitoring and control	High power consumption; frequent maintenance battery replacement)	Moderate robustness; performance degrades with interference and distance
Wi-Fi + 5G	Hybrid	Ultra-low latency varies from1–10 ms with 5G; supports real-time analytics	High energy consumption; higher infrastructure cost	High reliability in urban deployments; dependent on network coverage
SUN (Smart Utility Network)	Mesh	Moderate latency; suitable for scheduled data collection	Low-to-moderate power; designed for utility-scale deployment	High robustness in outdoor and industrial environments
Bluetooth	Star	Very low latency for short range; real-time possible locally	Very low power; minimal maintenance	Low robustness for large-scale deployment
XBee-PRO	Mesh / Point-to-Point	Low latency; supports real-time communication	Low-to-moderate power; stable maintenance requirements	High reliability in industrial environments; better range than Zigbee
Transceivers	Point-to-Point	Low latency; depends on implementation	Moderate power; maintenance depends on hardware design	Moderate reliability; sensitive to interference
Sigfox	Star	High latency; supports delayed transmission (not real-time)	Ultra-low power; very low maintenance	High coverage; robust for sparse deployments but limited bandwidth

encrypted data to the cloud, where RF, SVM, and XAI models are applied. In both cases, all ML/DL operations are executed in centralized cloud infrastructure.

- No study explicitly implements IoT + edge computing, where models are deployed on devices. Studies [11, 43], and [63] perform real-time monitoring and prediction; however, they do not specify the execution location of ML/DL models. Therefore, these cannot be classified as edge computing systems due to the absence of evidence for on-device or gateway-level inference.
- No study fully implements a hybrid edge–cloud architecture with defined task partitioning between edge and cloud. Studies [61, 73], and [95] exhibit partial characteristics, such as distributed sensing and preprocessing combined with ML.

However, none implement a pipeline where inference is split between edge and cloud components. Therefore, hybrid edge–cloud integration is not realized in the reviewed works.

5 Discussions

Across the reviewed literature, a clear evolution emerges in how WQP is being conceptualized, modelled and operationalized. Although the studies differ in their methodological choices, a shared foundation is evident, WQP increasingly depends on multi-parameter monitoring and analysis. Parameters such as pH, DO, temperature, BOD, COD, turbidity and TDS are repeatedly focused, these parameters are widely recognized as critical indicators of water potability, as they directly influence water's physical, chemical and biological characteristics. By examining several key indicators together researchers gain a more accurate understanding of the water's condition. This multi-parameter emphasis becomes the common thread linking traditional monitoring, ML/DL modelling and IoT-enabled sensing, but each domain contributes a distinct layer of capability and limitation.

Traditional methods of WQ testing remains the benchmark for accuracy involving lab and on-site evaluation, samples need to be gathered, transferred to laboratories and examined using certain tools and techniques. While they might be accurate, they are slow, labor-intensive and make real-time monitoring and swift response difficult, this inherent delay and resource demands highlight why computational approaches have gained prominence.

ML models represent the first major step toward scalable prediction. Best performing ML models in the reviewed papers are RF, GBR, XGBoost and PCA-QRF which delivered high predictive performance across mutli-parameter (DO, pH, BOD, COD, turbidity, TDS) with R^2 up to 0.94 and RMSE as low as 0.12. Optimized ensemble and hybrid models like AdCSO-sELM, Stacked RF, achieved high accuracy where water potability classification reached 96.54% accuracy and river water quality classification got 97.3% accuracy, highlighting improved prediction reliability across datasets.

DL approaches extend their capabilities by modelling complexity that ML models cannot easily capture. Hybrid and optimized models like GA-LSTM, FSGCN, ET-DFNN and Gated-LNN accurately predict key parameters (DO, WQI, BOD, TP) across diverse datasets. With $R^2 = 0.9995$, RMSE reductions up to 42%, and classification accuracy up to 99.74%. These models capture complicated temporal, spatial and nonlinear relationships in water quality data and perform better than traditional ML techniques.

IoT based systems add yet another dimension by transforming prediction from an analytical task into an operational capability. Sensors integrated with ML/DL models (RF, MLP, LSTM, SRU) enable continuous collections and accurate prediction of muti-parameters in WQ, achieving R^2 up to 0.99, MAE as low as 0.20 and classification accuracy > 95%.

- Integration of protocols like LoRa, Zigbee, MQTT, Wi-Fi and cloud processing allows rapid data transmission and analysis.
- IoT deployments shows that predictive modelling is no longer limited to offline analysis instead it can support timely water management interventions, improve prediction speed with advanced methods including XAI and QAOA optimization which help even in remote or harsh environments.

Additionally, IoT systems rely significantly on interpretability, data quality, and model robustness—areas where the methodological flaws found in ML/DL studies become especially significant. In operational situations, real-time predictions run the danger of being false in the absence of thorough validation and uncertainty reporting.

Taken together, the results from ML, DL and IoT domains point to a quickly developing field that is still developing in terms of methodological rigor. ML provides reliable baselines, DL pushes the boundaries of predictive accuracy and IoT enables real-time deployment, all three areas need more robust protections to guarantee that high reported performance translates into reliable, interpretable and generalizable systems. Machine learning, deep learning, and IoT enable real time water quality monitoring and early detection of contamination, supporting efficient resource use and informed decision making. These approaches enhance environmental resilience and contribute to SDG 6 Clean Water and Sanitation and SDG 11 Sustainable Cities and Communities.

5.1 Explainable AI techniques in water quality prediction

Recent studies highlight the growing importance of Explainable Artificial Intelligence (XAI) for improving the transparency and trustworthiness of ML and DL models in WQP. Techniques such as SHapley Additive exPlanations (SHAP) have been successfully employed to evaluate both global and local feature contributions, enabling validation of physical plausibility. For instance, SHAP-based analyses have revealing that COD and BOD are the most influential factors in determining water quality [99–101]. In [101] SHAP is applied as a post hoc interpretability tool. It ranks COD (6.92) and BOD (4.45) as predictors, establishes relationships through summary plots, and explains individual predictions using force plots. These outputs are not used for feature selection, model retraining, or decision-rule formulation, and no external validation is reported.

Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) plots further enhance interpretability by visualizing average and instance-specific feature effects. Several studies have demonstrated the utility of PDP for identifying optimal pH and temperature ranges, while ICE plots reveal heterogeneous responses across monitoring locations and seasonal conditions [102, 103].

Beyond interpretability, uncertainty quantification has emerged as a critical component of water quality forecasting. Ensemble-based models commonly report confidence intervals, whereas Bayesian approaches provide credible intervals that capture predictive uncertainty more comprehensively [102, 103]. Routine reporting of uncertainty intervals is therefore recommended to support risk-aware environmental management and regulatory decision-making.

In contrast, the chlorophyll-a prediction study [104] Jungsu Park et.al integrates SHAP into the model development pipeline. SHAP is used for feature ranking and selection from 18 variables, producing stable performance. The RMSE-observation standard deviation ratio (RSR) varies from 0.61–0.74, outperform feature importance (FI), and variance inflation factor (VIF) based selection. SHAP outputs are validated against observed data using target plots, and combined with PDP to identify quantitative thresholds varies from 17 to 23 °C. The study explicitly links SHAP results to sensor selection and monitoring design, reducing data collection cost. Here, XAI directly influences feature selection, model validation, and system-level decisions. In [105] SHAP is again used only after model training. It ranks features such as turbidity, solids, sulfate, hardness, and

pH, and explains prediction behavior through summary and local plots. However, these results are not used for feature reduction, model optimization, and no domain-level validation is performed.

Overall, [99] and [105] use XAI only for post hoc explanation, whereas Jungsu Park et.al [104] demonstrates integrated use where SHAP influences model design and operational decisions. None of the studies apply XAI within regulatory compliance frameworks or automated decision pipelines.

Among the reviewed works, [61] integrates XAI within an IoT–cloud framework using RF and SVM, where explainability is used to interpret nutrient prediction and support decision-making. In addition, detailed analysis of selected studies shows that [99] and [105] apply SHAP as a post hoc method to quantify feature importance and explain model predictions without influencing model design. In contrast, [104] uses SHAP in a functionally integrated manner: it performs feature ranking and selection, validates model behavior against observed data patterns, and combines SHAP with PDP to identify quantitative thresholds. No reviewed study applies XAI techniques such as LIME, attention mechanisms, or saliency maps.

With respect to uncertainty quantification, the reviewed literature shows limited implementation. Some studies employ ensemble models (e.g., [36, 53, 57]) that implicitly improve robustness and reduce prediction variance, but they do not report formal uncertainty measures such as prediction intervals or probabilistic outputs. None of the reviewed works implement Bayesian neural networks, Gaussian processes, or calibrated uncertainty estimation frameworks. As a result, uncertainty is not explicitly quantified or communicated to end users.

The practical role of XAI in the reviewed studies is twofold. First, it supports operational decision-making by identifying parameters which can guide sensor placement, monitoring priorities, and parameter control strategies. Second, in studies [104] XAI outputs are used to define data-driven thresholds and feature relevance, enabling operators to understand how changes in environmental conditions affect predictions. However, in most cases [99, 104] XAI remains descriptive, explaining model outputs without being integrated into automated decision pipelines.

Overall, the review establishes that the current state of the art is limited. XAI is predominantly implemented using SHAP-based feature attribution, with only one study [104] demonstrating integration into model design and deployment decisions. Uncertainty-aware learning is largely absent, with no study providing formal uncertainty quantification or reliability assessment. This indicates a clear research gap: future work must develop integrated XAI frameworks combined with rigorous uncertainty estimation, enabling trustworthy, regulation-ready WQP systems that can inform both operational control and policy-level decision-making.

5.2 Input parameter taxonomy and its impact on model performance

The reviewed studies use three categories of input parameters.

1. Minimal physicochemical parameters:
2. This category consists of pH, DO, BOD, COD, turbidity, and TDS, which are used in potability classification and WQI estimation studies [4, 36], and [53].
3. Extended chemical and nutrient parameters:

4. This category consists of Total Nitrogen (TN), Total Phosphorus (TP), Nitrate (NO_3^-), Ammonia (NH_3-N), and Phosphate (PO_4^{3-}), which are used in studies focused on eutrophication and nutrient dynamics, including [52] and [56].
5. Biological and advanced indicators:
6. This category consists of chlorophyll-a, E. coli, and organic carbon, and inputs like satellite-derived features and spectral data, as used in [37, 50], and [41].

These categories correspond directly to the prediction task, with minimal sets used for classification, extended sets for nutrient prediction, and hybrid inputs for spatial and ecological modeling. The choice of input parameters has a measurable effect on model performance. Studies using minimal parameter sets achieve high accuracy in datasets, such as 62.21% accuracy in potability prediction [4] and up to 97.3% accuracy in classification tasks [36], but these models rely on limited variables. Studies incorporating nutrient parameters report improved performance in predicting variables such as Chl-a, TN, and TP, with R^2 values of 0.823, 0.619, and 0.691 respectively [52]. Spatiotemporal studies using extended parameter sets reduce prediction error, with MAE reductions of 32–46% [56]. Models using satellite and hyperspectral inputs improve spatial prediction accuracy, achieving R^2 of 0.87 [37]. These results show that increasing parameter diversity improves prediction accuracy for complex environmental processes, while also increasing data requirements and model complexity.

Feature importance and selection analyses identify task-specific dominant parameters. In WQI prediction studies, tree-based models and SHAP analysis rank COD and BOD as the most influential variables, followed by DO, while EC, chloride, and nitrate show minimal contribution [99]. In potability classification, SHAP-based analysis identifies pH, hardness, turbidity, sulfate, and total dissolved solids as key predictors [105]. In chlorophyll-a prediction, variables such as upstream Chl-a concentration, temperature, and conductivity show the highest contribution to model output [104]. Studies that apply SHAP for feature selection demonstrate that reducing input variables based on SHAP rankings maintains model performance while lowering data collection requirements [104]. These results establish a direct link between feature importance analysis and sensor selection, where parameter selection must be aligned with the prediction objective to ensure both accuracy and deployment efficiency.

5.3 Methodological quality assessment and critical interpretation of results

The methodological quality assessment conducted on the 45 studies with sufficient extractable information indicates that the majority of works follow standard experimental practices in ML/DL-based WQP research. All reviewed studies reported performance metrics, and most employed systematic validation approaches, with 97.8% evaluating multiple algorithms or baseline models. Many studies utilized k-fold cross-validation along with multiple evaluation metrics such as accuracy, chi-square, and R^2, supported by well-defined training setups, experimental procedures, and data usage. While explicit reporting of data leakage prevention and external validation was limited, no study indicated the presence of such issues. Overall, the reported high predictive performances can be attributed to structured experimentation and effective model design, although their generalizability across diverse spatiotemporal settings would benefit from further validation. To further strengthen the reliability and applicability of ML/DL-based

WQP systems, future research should emphasize clearer documentation of validation strategies, including temporal and cross-location evaluation.

To improve the dependability and usefulness of ML/DL-based WQP systems and to guarantee realistic generalization in practical applications, future research should give precise documentation of preprocessing operations, leakage-avoidance techniques, cross-location validation and site-held-out validation and temporal or geographical splits.

The study [106–111] reviews recent forecasting studies that utilize machine learning and deep learning models for WQP across diverse environmental settings.

5.4 Cross-site transfer and domain adaptation

Various study examines a model's capacity to generalize outside its site and conditions to assess external validity and transfer in WQP. Cross-site transfer tests, such as training on river data and testing on lake data, show that broad alterations in site attributes and climatic circumstances can reduce direct transfer accuracy from 90 to 72%. Basic baseline evaluations employing train-here/test-there deltas quantify this performance gap. Domain adaptation options include fine-tuning pre-trained models with 10–15% of target site data and feature normalization based on hydrological seasons like monsoon, dry, and post-monsoon to improve transferability. These methods limit seasonal and climatic distribution fluctuations, boosting model resilience. Cross-site transfer studies, baseline comparisons, and domain adaptation processes improve WQP models' external validity across varied water bodies and climatic locations. Applicable to various waterways and climates. A model or methodology's ability to generalize beyond its original site and conditions is called external validity and transfer. Mu et al. [112] showed that deep representation learning enables cross-basin WQP under data-scarce conditions by learning transferable features, demonstrating strong external validity across different basins. Mu et al. [113] emphasized cross-site transfer and domain adaptation in WQP by evaluating river-to-lake and temperate-to-tropical transfers using simple train-here/test-there baselines, and by improving generalization through fine-tuning and feature normalization based on hydrological seasons.

5.5 Interplay between fair data principles and the EU AI act

Beyond model accuracy, robust deployment of ML, DL, and IoT-enabled WQP systems require strong data governance, lifecycle management, and regulatory alignment. Data used for environmental monitoring should adhere to the FAIR principles Findability, Accessibility, Interoperability, and Reusability to ensure transparency, reproducibility, and long-term usability across institutions and jurisdictions [114]. From a model lifecycle perspective, operational systems must incorporate continuous monitoring, data and concept drift detection, and clearly defined re-training cadences to account for seasonal variability, sensor degradation, infrastructure changes, and evolving anthropogenic pressures. Regulatory and organizational touchpoints are equally critical, particularly for public utilities and environmental agencies, where explainability is essential for public decision-making, regulatory reporting, and stakeholder trust, while procurement and maintenance constraints often limit the feasibility of highly complex or opaque models. In the European context, emerging frameworks such as the EU AI Act adopt a risk-based approach that may classify environmental monitoring and decision-support systems as

requiring enhanced transparency, data quality assurance, and human oversight [115]. Consequently, future WQP frameworks should be designed not only for predictive performance but also for governance readiness, lifecycle robustness, and regulatory compliance to support responsible, trustworthy, and sustainable real-world deployment.

5.6 Comparative analysis of Ml and DL models for water quality prediction

Under similar dataset conditions, model performance is determined by data structure and size.

1. For datasets having ≤ 2000 samples, tree-based ML models like DT, RF, and XGBoost achieve the highest accuracy, with DT and RF reaching up to 99.8% accuracy in TDS prediction [75] and XGBoost shows reliable performance on datasets having 29 samples [48].
2. For medium-scale datasets with multiple physicochemical parameters, ensemble models like PCA-QRF and soft voting classifiers provide superior performance, achieving R^2 values up to 0.94 and accuracy up to 99.91% [53, 57].
3. In time-series datasets involving temporal dependencies, such as river and aquaculture systems, DL models like LSTM, CNN-LSTM, and GRU outperform ML models by reducing prediction error. The CNN-LSTM achieve RMSE as 0.0827 [41] and GA-LSTM reducing RMSE by 42.42% [42].
4. For spatiotemporal and graph-structured datasets, graph-based DL models like FSGCN, MDMVL, and DRL-GCN achieve the best results, with MAE reductions of 32–46% and correlation up to 0.928 [55, 56].
5. In high-dimensional datasets derived from hyperspectral or satellite data, hybrid approaches combining CNN/RNN with ML models such as RF and Gradient Boosting provide the highest performance [50].

5.7 Performance and cost analysis

1. DL models outperform ML only when the dataset includes temporal sequences, spatial relationships, or high-dimensional inputs.
2. In small or medium datasets with input parameters, ML models exceed DL performance. This performance gain in DL comes at a higher cost: DL models require large datasets, longer training time, GPU-based computation, and extensive hyperparameter tuning.
3. In contrast, ML models require less data, train faster, and can be deployed on low-resource IoT systems. Therefore, the selection of DL over ML must be justified by the presence of temporal or spatial complexity and the availability of sufficient computational resources.

5.8 Trade-off between accuracy, interpretability, and implementation complexity

The trade-off between accuracy, interpretability, and implementation complexity is explicit across model categories. Tree-based ML models provide high accuracy with full interpretability through feature importance and decision rules, and they have low implementation complexity. Ensemble ML methods further improve accuracy but reduce interpretability due to model aggregation, with moderate implementation complexity. DL models achieve the highest accuracy in large datasets but act as black-box

models with low interpretability and high computational complexity. Graph-based and hybrid DL models further increase accuracy in spatiotemporal systems but require complex data preprocessing and architecture design. As a result, ML models are suitable for interpretable, low-cost deployments, while DL and hybrid models are appropriate for high-accuracy requirements in large-scale and dynamic environments.

5.9 Generalizability, transferability, and regional dependence

WQPs have made significant progress with the integration of IOT and intelligent systems. Still, generalizability in different regions is a common challenging issue. In recent models [5], models are trained with the special data set, i.e., based on region-specific, constrained with diverse environmental conditions. This problem occurs due to the climate and ecological components having a significant role in WQPs.

The current research is based on a single, specific region-based dataset that directly impacts the robustness of the system. In [4, 13], and [86], the high accuracy of the machine learning model is shown, but it isn't validated with a diverse dataset. Also, different enhanced deep learning models, such as LSTM and ensemble approaches, define high prediction accuracy [39, 49, 74], but their limited cross-regional validation is identified. Peerzada and Kamat's [49] water quality model faces an issue of accuracy with the diversity of the dataset.

To overcome the existing issue, i.e., the diversity of the dataset, transfer learning and domain adaptation approaches are utilized. In papers [112, 113] Transnet is utilized to transfer the knowledge between regional datasets. Also, this approach enhanced the accuracy with a limited dataset. Additionally, IoT and ML models, i.e., a hybrid model, allow the real-time data from different regions that directly impact efficient transfer learning. Explainable AI models [99, 100] enhance the interpretability of the model to adapt to the different region-based conditions. Many factors directly impact transferability, such as changes in climate [17], inconsistency of the sensor [12], and different land pollution sources [27]. For the generalization of the model, the different benchmarks of the WQI parameters [31], the standard dataset, and the inclusion of pH and water flow [34].

According to the existing literature review, various best practice methods can optimize the performance of the system through a cross-regional dataset. First, utilize the hybrid dataset that captures the diverse dataset and enhances the robustness of the system. For a small-sized dataset, utilize the transfer learning approach [104]. Hybrid and multimodal machine learning enhance the accuracy of the system [36, 53]. The concatenation of IoT applications provides high performance on real-time datasets [62]. Explainable AI methods provided the transparency and trust of the model [99].

5.10 Challenges

As an answer to RQ4, this section outlines the key challenges. While there has been considerable advancement in predictive models [116, 117] for forecasting water quality, certain challenges persist. The recognized challenges (see Fig. 9) such as those arising from rapid urbanization, industrial activities and environmental disturbances like erosion, hurricanes and floods highlight the need for reliable prediction models [118].

Some limitations include:

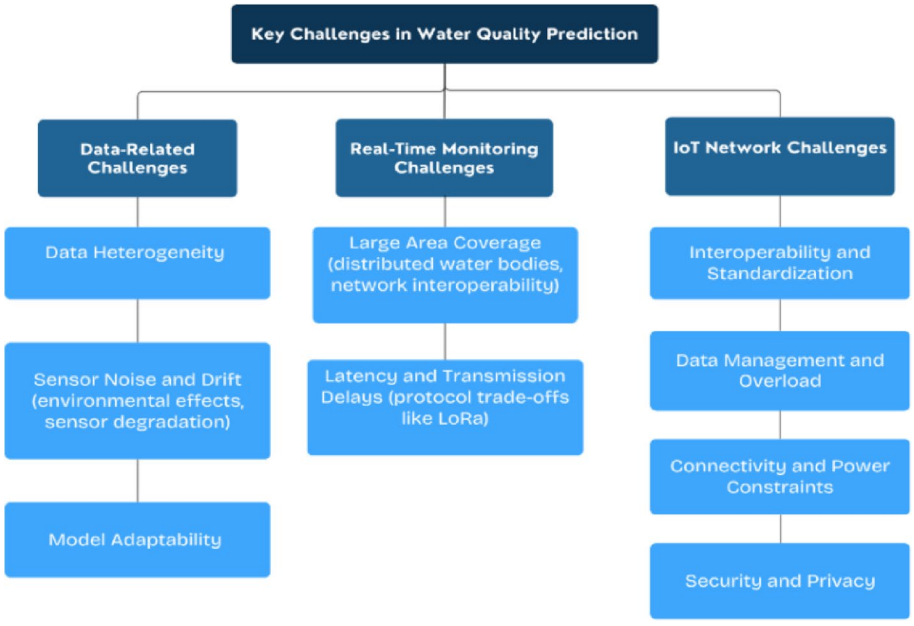


Fig. 9 Challenges in Water Quality Prediction

1) Data Heterogeneity: Sensor data comes in diverse formats, resolutions and time intervals. Combining this into consistent, structured input for ML/DL models requires intensive pre-processing.
2) Sensor Noise and Drift: Changes in weather, dirt buildup, or unstable power can make sensors give wrong readings over time. If not cleaned or adjusted regularly, the data becomes less accurate. Filters like Kalman filters can help fix this.
3) Real-time monitoring over large areas: Rivers and lakes span a large geographical area and are often connected through various networks for data sharing and real-time prediction. Establishing a seamless and integrated system for such large-scale real-time monitoring is challenging due to the because of how water bodies are distributed in location and the need for consistent communication across diverse network infrastructures
4) Latency and Delay in Transmission: Real-time systems face latency issues in rural or remote locations. Protocols like LoRa offer long-range capabilities but trade off data rate and timeliness.
5) Streaming Model Adaptability: Most ML/DL models are trained offline and struggle to adapt continuously to streaming or evolving data. Integrating online learning, transfer learning or model fine-tuning mechanisms is crucial for dynamic water environments.

The review does not include a formal quality assessment or risk-of-bias evaluation of the selected studies. Consequently, all studies were considered without weighting for differences in dataset size, validation strategy, or reporting standards. This directly affects the comparability of reported performance metrics, as variations in experimental design are not controlled. Therefore, the model performance comparisons, rankings, and trends presented in this review are based on reported results from the original studies without adjustment for study quality.

Challenges Faced in IoT Network Systems for Water Quality Monitoring:

1. Interoperability and Standardization: Heterogeneous sensors, communication protocols, and platforms are frequently used in IoT water quality monitoring systems. The absence of widely accepted standards makes it challenging to create scalable, multi-vendor networks by impeding smooth integration and data interchange. System maintenance may become more difficult as a result of this fragmentation, which may lead to inconsistent data formats.
2. Data Management and Overload: Dispersed sensors in IoT deployments produce enormous amounts of continuous data streams. Strong infrastructure and analytics skills are necessary for the effective processing, analysis and storage of this large data. It becomes difficult to extract useful insights from the stream of unprocessed sensor data without efficient data management.
3. Connectivity and Power Limitations: A large number of water monitoring locations are situated in isolated or subterranean areas with poor network coverage. Real-time data transmission is impacted by the challenge of maintaining dependable wireless access. Furthermore, it is still difficult to power sensor nodes for prolonged periods of time, particularly in situations where changing batteries or connecting to power grids is not feasible.
4. Data Security and Privacy: Wireless data transmission over Zigbee, MQTT or Wi-Fi introduces risks of interception. Secure communication protocols, encryption layers, and blockchain-based logging may safeguard sensitive environmental data.

5.11 Future scope

Effectively addressing the challenges outlined in the previous section sections for innovative solutions that involve emerging technologies and sustainable practices.

For instance, ML models can assess large and complex datasets on WQ with high prediction accuracy and help in identifying trends and patterns [23]. Sharing data across remote locations requires resources, funding and infrastructure. However, the transmission of data streams from geographically dispersed and remote sensor networks to centralized cloud required bandwidth constraints.

Edge AI enables local data processing at the network edge, reducing latency and energy consumption while improving the efficiency and adaptability of water quality monitoring systems.

Federated Learning (FL) preserves privacy by enabling collaborative model training without sharing raw data. Its wider use can improve system resilience while addressing the increasing demands for data security. Explainable AI increases prediction interpretability and transparency, which promotes regulatory acceptability and trust. Hybrid physics ML models combine traditional physics models with ML models to improve accuracy and predictions across diverse environmental conditions in the future.

Hybrid physics-informed machine learning (PIML) models improve consistency and accuracy by integrating physical processes into data-driven frameworks for WQP. Moeini and Abokifa [118] created a physics-informed ensemble ML framework that embeds governing Partial Differential Equations (PDEs) advection, dispersion, and response dynamics to improve predictive performance over data-driven models. These hybrids can include mass-balance constraints to conserve pollutants and monotonicity constraints (e.g., dilution with increasing flow or temperature-dependent reactions) for physically plausible and interpretable WQPs.

Furthermore, the dynamic nature of environmental pollutants and hydrological conditions requires ongoing model and domain adaptation strategies to sustain generalizability and predictive reliability. Benefiting from these models at a large scale demands robust infrastructure to share data across vast and often remote water bodies. Investment in resources, funding and interoperable systems are essential to enable real-time insights and ensure effective monitoring.

The high-quality research, and the diversity of metrics and evaluation protocols highlights the breadth of approaches, offering a comprehensive perspective and supporting reproducibility for future studies.

6 Conclusion

This systematic review was conducted in accordance with PRISMA guidelines to comprehensively analyze recent advancements in ML, DL, and IoT based approaches for WQP and monitoring. The review process involved systematic identification, screening, and evaluation of peer-reviewed studies from selected scientific databases, ensuring methodological rigor and relevance. The included studies utilized datasets collected from different geographical regions and environmental conditions, highlighting the growing global interest in intelligent water quality assessment systems.

The review demonstrates that ML-based models, particularly ensemble approaches such as Random Forest and Gradient Boosting, achieve strong predictive performance in WQP tasks, with several studies reporting accuracies exceeding 97%. Their effectiveness is largely attributed to robust feature learning and adaptability to heterogeneous environmental datasets. Similarly, DL models including CNN-LSTM, attention-based architectures, and hybrid frameworks further enhance predictive capability by effectively capturing complex spatial and temporal dependencies, with many studies reporting R^2 values greater than 0.90.

Real-world IoT deployments for water monitoring vary by communication technology. Long-range technologies such as LoRa/LoRaWAN and Sigfox provide extended communication coverage, high reliability, and low power consumption, making them suitable for large-scale remote monitoring applications. In contrast, Zigbee and mesh-based networks offer low-latency and reliable multi-hop communication for localized deployments, although they are constrained by shorter transmission ranges and higher maintenance requirements.

Overall, the integration of ML, DL, and IoT technologies presents significant potential for the development of intelligent and automated water quality management systems. Future research should emphasize scalable, energy-efficient, and edge-enabled architectures with robust validation frameworks to support reliable and sustainable real-world WQP applications.

Abbreviations

AN	Ammonium nitrogen
AR	Autoregressive
BOD	Biochemical oxygen demand
Ca	Calcium
CDOM	Colored dissolved organic matter
CNN	Convolutional neural networks
Chl-a	Chlorophyll-a
COD	Chemical oxygen demand
COND / EC	Electrical conductivity
DL	Deep learning
DIN	Dissolved inorganic nitrogen

DO	Dissolved oxygen
DS	Dissolved solids
DT	Decision tree
E. coli	Escherichia coli
EC	Electrical conductivity
FSA	Fuzzy similarity analysis
FA	Factor analysis
GB / GBM	Gradient boosting / Gradient boosting machine
IOT	Internet of things
K	Potassium
KNN	K-Nearest neighbor
LR	Logistic regression
ML	Machine learning
MAE	Mean absolute error
MAPE	Mean absolute percentage error
Mn	Manganese
MODIS	Moderate resolution imaging spectroradiometer
NARNet	Nonlinear autoregressive network
NB	Naïve Bayes
NH_3–N/NH_4–N	Ammonia Nitrogen/Ammonium Nitrogen
Ni	Nickel
NO_3	Nitrate
NO_3-N	Nitrate nitrogen
NTU	Nephelometric turbidity units
NSE	Nash–Sutcliffe efficiency
ORP	Oxidation–reduction potential
P	Phosphorus
PCU	Platinum-cobalt units
PCA	Principal component analysis
PAR	Photosynthetically active radiation
PDR	Packet delivery rate
PL	Path loss
QRF	Quantile regression forest
RF	Random forest
RNN	Recurrent neural networks
RMSE	Root mean squared error
R^2	Coefficient of determination
SAL	Salinity
SAR	Sodium adsorption ratio
SC	Specific conductance
SVM	Support vector machine
SVR	Support vector regression
SWM	Sinusoidal weighted mean
SRU	Simple recurrent unit
TEMP	Temperature
THMs	Trihalomethanes
TP	Total phosphorus
TDS	Total dissolved solids
TSS	Total suspended solids
TN	Total nitrogen
TUR	Turbidity
WQI	Water quality index
WQC	Water quality classification

Author contributions
Katyayani Pandey contributed to the design and development of the research methodology, data acquisition and validation of results. Dr.Praveen Kumar Shukla supervised the research, provided critical insights for methodological refinement, and contributed to manuscript preparation and final review. Dr. Suman Bhakar provided valuable mentoring support and assisted in the review and refinement of the manuscript. Dr.Hitesh Tekchandani offered constructive feedback and Dr. Rohit Mittal offered valuable guidance during the review process.

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